

A 33 yo M with a history of a kidney transplant (CMV donor +/- recipient +/-) 6 months ago for IgA nephropathy is found to have a Cr of 3.4 (prior Cr 1.5). He is on tacrolimus, prednisone, and mycophenolate.

Click here

What is your differential diagnosis?

- Etiologies can be classified by:
- 1) Pre-, intra-, and post-renal causes
 - 2) [Timing from transplant](#)

Click here

	PRE-RENAL	INTRA-RENAL	POST-RENAL
Typical/ Non-transplant specific	- Sepsis - Medications - Differential? - Volume depletion	- Acute tubular necrosis - Contrast nephropathy - Differential? - nephritis	- Obstruction from 3C's (Calculi, Clot, Cancer) - Differential?
Transplant specific	- Acute calcineurin - Differential? - thrombosis	- Acute rejection - Differential? - Chronic calcineurin inhibitor toxicity	- Ureteral stenosis - Differential? - hematoma, urinoma



How does the time from transplant affect our differential diagnosis?



Renal duplex, CNI trough, biopsy, viral studies



PRE-RENAL

Renal artery/ vein thrombosis

Vascular thrombosis
Renal duplex

Renal artery stenosis
Renal duplex

Renal artery stenosis

Acute CNI toxicity
Neuro/nephrotoxic, HTN
↑ K, ↑ uric acid
↑ CNI trough level

Acute calcineurin inhibitor toxicity

INTRA-RENAL

Hyperacute rejection

Hyperacute rejection (< 24 h)
DSAs

Acute rejection

Acute rejection
Graft pain, fever
Proteinuria (especially, > 1g/d)
• Acute cellular (T cell)– biopsy
• Acute humoral (antibody) – biopsy, DSAs

Glomerulonephritis (recurrence or de novo)

BK nephropathy

BK nephropathy
Hemorrhagic cystitis
Serum BK PCR (high Sn/Sp), Urine BK PCR, biopsy

Can cause extrinsic compression
Renal US

CMV nephritis
↑ risk in CMV D+/R-
CMV PCR, biopsy

CMV nephritis

POST-RENAL

Lymphocele
Perinephric hematoma
Urinoma

Ureteral stenosis

Ureteral stenosis
Renal US



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Case 1

ROS: new numbness/ tingling in his hands/feet, pain and redness in his L 1st metatarsal.

Meds: fluconazole 2 weeks ago for a yeast infection. No other med changes.

Exam: BP 160/ 92, other VS normal. No tenderness over his graft.

135		52	}	FeNa - 0.7%
6.9		3.4		

UA – clear with hyaline casts, no protein, WBCs, or RBCs



Cr improved to 3.1 with 1 L IVF.

CNI trough elevated at 14, goal 6-8.

Normal arterial flow, no hydronephrosis.

Not done.

Urine BK PCR level elevated at 14k.

Acute CNI toxicity resulting in neurotoxicity, gout flare, ↑ BP, ↑ K. Has underlying BK nephropathy. Fluconazole interacts with CNIs to ↑ CNI levels.

Case 2

ROS: fevers/chills for 2 days, abdominal pain, dysuria, nausea/vomiting.

Exam: T 39.4C, HR 115, BP 160/ 92. Tenderness over his graft.

135		32	}	FeNa - 1.2% Urine Pr:Cr - 2.6
3.5		3.4		

UA – cloudy, + leukocyte esterase, + nitrite + WBCs, + RBCs, + protein, + bacteria.



Cr improved to 3.1 with 1 L IVF.

CNI trough within range at 7.

Normal arterial flow, no hydronephrosis. Perinephric stranding.

Acute cellular rejection. No DSAs.

Urine BK/adeno PCR negative. CMV PCR negative.

Acute cellular rejection and sepsis from acute graft pyelonephritis.

Case 3

Renal biopsy yesterday for uprending Cr, complicated by hematuria.

ROS: acute onset suprapubic/abdominal pain, no urine output for 12 hours

Exam: VSS. Abdomen tender over graft/suprapubic region.

135		32	}	FeNa - 5.7% Urine Pr:Cr - 0.6
3.5		3.4		

UA – none collected



Cr unchanged at 3.3 after 1 L IVF.

CNI trough within range at 7.

Normal arterial flow, graft hydronephrosis. Bladder with 700 cc of urine and retained clot.

Glomerulosclerosis and interstitial fibrosis concerning for chronic CNI toxicity.

Not sent.

Acute clot retention resulting in obstructive uropathy on background of chronic CNI toxicity.

Allograft monitoring has moved toward earlier, less-invasive detection:

Plasma (serum) BK PCR is the preferred screening/diagnostic test for BK — monthly until ~9 mo post-transplant, then q3mo to 2 yr (2nd Int'l Consensus, Transplantation 2024). Urine BK PCR/decoy cells are a sensitive early screen; sustained plasma DNAemia (>1,000 copies/mL) or biopsy-proven BKVAN → reduce immunosuppression.

Donor-derived cell-free DNA (dd-cfDNA) supports — does not replace — biopsy: use to help exclude rejection (esp. antibody-mediated) in acute allograft dysfunction or stable function (ASTS statement, Oct 2024). Biopsy + DSAs remain the diagnostic standard.

CMV reactivation should prompt active BK screening (CMV-BK coinfection raises graft-loss risk).

Core teaching unchanged: 5 transplant-specific causes + timeline-driven differential + renal duplex / viral PCR / CNI trough / biopsy toolbox.

Take-Home Points

1. Keep BOTH lists in your differential — transplant-specific AND usual AKI causes (this patient still gets pre/intra/post-renal).
2. Five transplant-specific causes: CNI toxicity (acute + chronic), renal artery stenosis/thrombosis, viral (BK, CMV, adenovirus), rejection, ureteral stenosis.
3. Timing narrows it: <24 h hyperacute rejection / vascular thrombosis; first 3 mo acute CNI toxicity & rejection; 3 mo–2 yr renal artery stenosis, BK (peak ~10–13 mo), CMV.
4. Toolbox: renal duplex US, plasma + urine BK/CMV PCR, CNI trough, allograft biopsy + DSAs ($\pm$ dd-cfDNA). Loop in transplant nephrology early.
5. Don't forget drug interactions — azole antifungals raise CNI levels and can precipitate acute CNI toxicity.

Attribution & Changelog

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Corrections (surgical, animations preserved): “mycophenylate” → “mycophenolate”; “tranpslant” → “transplant”; “viriuria” → “viruria”.

Added: 2024 practice-update slide (plasma BK PCR screening; dd-cfDNA role) — standing teaching unchanged.

Companion files: presenter board + learner board (both corrected). Educational use only; clinician-review medical content before teaching.